PTVA's

MULUND COLLEGE OF COMMERCE (AUTONOMOUS)

Programme: B.Sc. Information Technology **Class: TY** **Semester: V**

Exam: ESE **Name of the Course: Internet of Things**

Date of Exam: 10/10/2025 **Marks: 60** **Time: 2 hours**

Instructions:

(1) **All** questions are **compulsory**.
(2) Make **suitable assumptions** wherever necessary and **state the assumptions** made.
(3) Answers to the **same question** must be **written together**.
(4) Numbers to the **right** indicate **marks**.
(5) Draw **neat labeled diagrams** wherever **necessary**.
(6) Use of non-programmable calculator is **allowed**.

Q 1 Attempt *any TWO* of the following: **12**

A. Illustrate the 6LoWPAN Architecture.

B. Imagine a "smart" chair as an IoT device. Describe three potential data inputs (via sensors) and one possible response (via actuators) it could have and explain how this creates a two-way interaction.

C. Distinguish between TCP and UDP.

D. Evaluate the claim that "IoT is changing how we live, work and interact with the world" by discussing its potential benefits and challenges.

Q 2 Attempt *any TWO* of the following: **12**

A. What are the core components typically involved in an IoT prototype?

B. Identify the types of sensors and describe how they can be applied in a smart home embedded system project.

C. How do technical and manufacturing factors influence the shift from a prototype to a production-ready IoT device?

D. Evaluate the necessity of having an operating system on an SoC for multitasking applications.

Q 3 Attempt *any TWO* of the following: **12**

A. Determine the features that need to be considered while choosing a laser cutter.

B. Explain the following protocols.

(i) Message queuing telemetry transport protocol.
(ii) Extensible messaging and presence protocol.

C. Differentiate between Public, Partner and Private APIs with suitable IoT-related examples.

D. Justify the importance of starting with non-digital prototyping methods in product design with an example.

Q 4 Attempt *any TWO* of the following: **12**

A. List the different types of memory used in embedded systems and mention feature of each.

B. Describe the various types of business models that can be adopted by IoT companies.

C. Compare Desktop debugging and Embedded debugging.

D. Assess the effectiveness of different venture capital exit strategies in the context of IoT startups.

Q 5 Attempt *any TWO* of the following: **12**

A. Illustrate the process of transforming a prototype into a manufacturable electronics product.

B. Apply the role of technology in control, highlighting its benefits and drawbacks with suitable examples.

C. Summarize the process of designing a PCB from schematic to testing.

D. Evaluate the impact of IoT devices on personal privacy and suggest measures to mitigate privacy risks.